Machine Learning

Gaussian Mixture Model & EM

Dong-Kyu Chae

Assistant Professor, Dept. of Computer and Software

Hanyang University

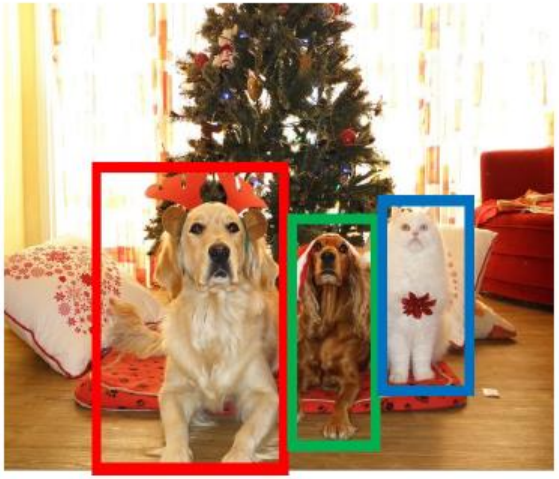
Supervised VS. Unsupervised

Supervised learning

- Data (**x**, **y**)
  - **x** is data, **y** is **label**
- Goal
  - Learn function to map **x** -> **y**
  - Performing specific task
- Examples

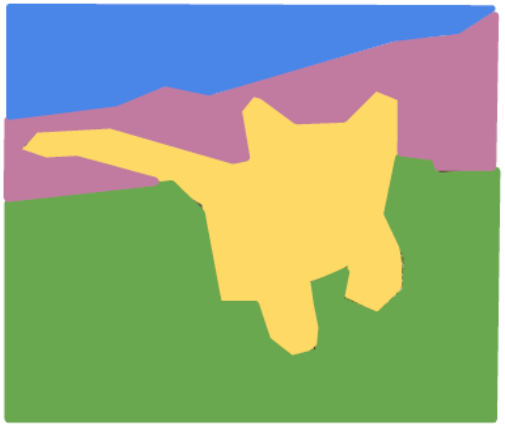
Cat

Classification



DOG, DOG, CAT

Object Detection



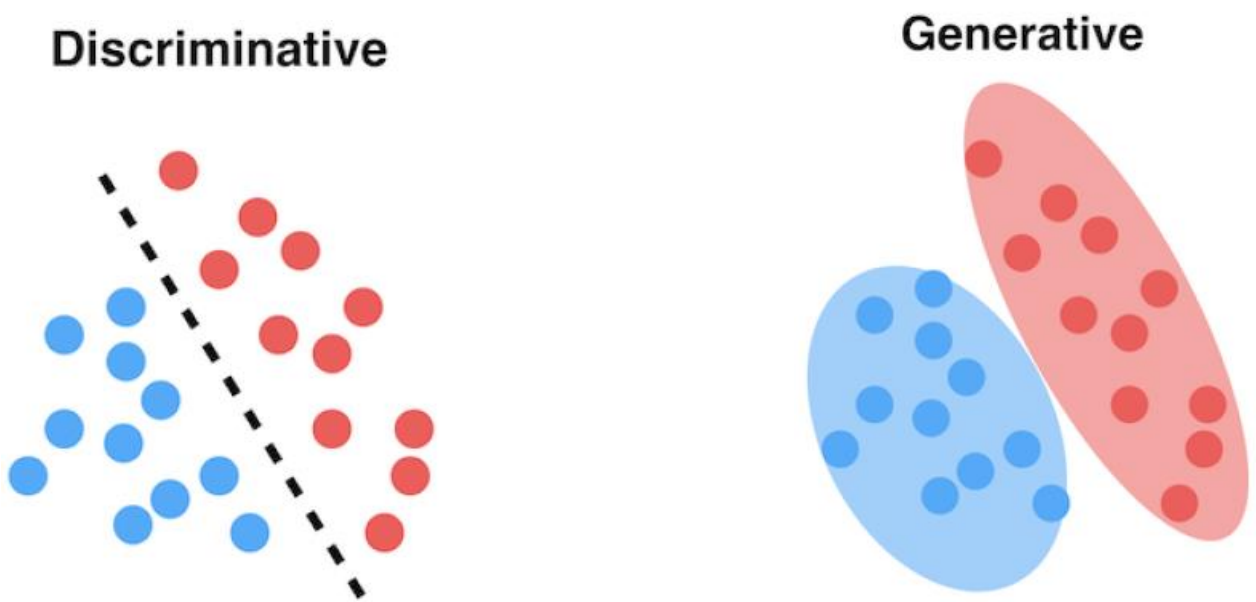
GRASS, CAT, TREE, SKY

Semantic Segmentation

Supervised VS. Unsupervised

Unsupervised learning

- Data (**x**)
  - It focuses on data x itself
  - Label **y** is not mandatory
- Goal
  - Learn a model that represents **data distribution** underlying **x**
  - Learn the internal structure of data itself



Today's Contents

- ☐ Latent Variable Models (LVM)
- ☐ Gaussian Mixture Models (GMM)
- ☐ K-Means Clustering
- ☐ Expectation Maximization (EM)

Latent Variable Models

- **Observed data x** might be influenced by, or originated from **latent variables z**

Myth of the cave

Introducing Latent Variable z

- **Recall our objective:**
  - Find $\hat{\theta} = \underset{\theta}{\operatorname{argmax}} \ln p(\mathbf{x} \mid \theta)$

- **By the sum rule:**

$$\ln p(\mathbf{x} \mid \theta) = \ln \sum_{\mathbf{z}} p(\mathbf{x}, \mathbf{z} \mid \theta)$$

  - $\mathbf{z}$: ***latent*** variable(s), which is a hidden common "cause" for observed variables

- **Why do we introduce the random variable(s) z?**
  - There might be ***hidden*** observations/***missing*** values ⇒ let z represent them
  - Sometimes $\ln p(x|\theta)$ is difficult to optimize directly, but $\ln p(\mathbf{x}, \mathbf{z}|\theta)$ may be easy to handle

Example: A Coin-Flipping Experiment

- **Flipping process**
  - We have two coins
  - Randomly (50% : 50%) choose one of the two coins
  - Perform ten independent coin tosses with the selected coin
  - Thus, the entire procedure involves a total of **50** tosses
- **What is probability of heads? (to estimate $\hat{\theta}_A$ and $\hat{\theta}_B$)**

B	H	T	T	T	H	H	T	H	T	H
A	H	H	H	H	T	H	H	H	H	H
A	H	T	H	H	H	H	H	H	T	H
B	H	T	H	T	T	T	H	H	T	T
A	T	H	H	H	T	H	H	H	T	H

What if observed? Or unobserved?

Preliminary

- **Two types of data: $\mathbf{x} = \{x_1, \dots, x_5\}$ and $\mathbf{z} = \{z_1, \dots, z_5\}$**
  - $x_i \in \{0,1, \dots, 10\}$: ***# heads observed***
  - $z_i \in \{A, B\}$: ***identity*** of coin used (hidden or not)

- **Probability of heads**
  - Coin A: $\binom{H_A+T_A}{H_A} \hat{\theta}_A^{H_A} (1 - \hat{\theta}_A)^{T_A}$
    - Coin A lands on **Heads** with probability $\hat{\theta}_A$ and Tails with $(1 - \hat{\theta}_A)$
  - Coin B: $\binom{H_B+T_B}{H_B} \hat{\theta}_B^{H_B} (1 - \hat{\theta}_B)^{T_B}$
    - Coin B lands on **Heads** with probability $\hat{\theta}_B$ and Tails with $(1 - \hat{\theta}_B)$

- **Goal: to estimate $\hat{\theta}_A$ and $\hat{\theta}_B$**
  - By MLE

Case 1: Coin Observed

Maximum likelihood estimation of $\boldsymbol{\theta}$

- Coin A: $\prod_A p(x_n \mid \hat{\theta}_A) = \hat{\theta}_A^{H_A}(1-\hat{\theta}_A)^{T_A}$
- Coin B: $\prod_B p(x_n \mid \hat{\theta}_B) = \hat{\theta}_B^{H_B}(1-\hat{\theta}_B)^{T_B}$

$$\underset{\hat{\theta}_A}{\operatorname{argmax}} \ln\{\hat{\theta}_A^{H_A}(1-\hat{\theta}_A)^{T_A}\} = \underset{\theta}{\operatorname{argmax}}\, H_{\mathrm{A}}\ln\hat{\theta}_A + T_{\mathrm{A}}\ln(1-\hat{\theta}_A)$$

$$\frac{d}{d\hat{\theta}_A}(H_{\mathrm{A}}\ln\hat{\theta}_A + T_{\mathrm{A}}\ln(1-\hat{\theta}_A)) = 0 \quad \Rightarrow \quad \boxed{\hat{\theta}_A = \frac{H_A}{H_A+T_A}, \qquad \hat{\theta}_B = \frac{H_B}{H_B+T_B}}$$

B	H	T	T	T	H	H	T	H	T	H	H: 5 T: 5
A	H	H	H	H	T	H	H	H	H	H	H: 9 T: 1
A	H	T	H	H	H	H	H	H	T	H	H: 8 T: 2
B	H	T	H	T	T	T	H	H	T	T	H: 4 T: 6
A	T	H	H	H	T	H	H	H	T	H	H: 7 T: 3

$$\hat{\theta}_A = \frac{24}{24+6} = 0.80$$

$$\hat{\theta}_B = \frac{9}{9+11} = 0.45$$

Case 2: Coin Unobserved

- **Now let's assume that the variable $\mathbf{z}$ is not visible**
  - Given: $\mathbf{x} = \{x_1, \dots, x_5\}$ where each $x_i \in \{0,1, \dots, 10\}$
  - ***Hidden***: $\mathbf{z}$ (*identities* of coins used for each set of tosses)
  - We derive $\mathbf{z}$ as our ***hidden variables*** or ***latent factors***

- **We still wan to estimate $\hat{\theta}_A$ and $\hat{\theta}_B$**

- **A naïve approach: guess, and update**
  - *Guess* which coin was used in each of five sets by somehow,
  - And then *complete* the hidden data based on our guess
  - Then, we do MLE with the completed data

- ***Iterate the process until converge***

Case 2: Coin Unobserved

- **Starting from some initial parameters** $\hat{\theta}^t = \{\hat{\theta}_A^t, \hat{\theta}_B^t\}$
- **One iteration :**
  1. Guess (for each of the 5 sets) whether coin A or coin B was more likely to have generated observed flips (based on current parameter estimates)
  2. Assume theses guessed coin assignments to be correct and apply regular MLE procedure to get $\hat{\theta}^{t+1}$
- **Repeat theses two steps until convergence**

Case 2: Coin Unobserved

Example:

B	H	T	T	T	H	H	T	H	T	H
A	H	H	H	H	T	H	H	H	H	H
A	H	T	H	H	H	H	H	H	T	H
B	H	T	H	T	T	T	H	H	T	T
A	T	H	H	H	T	H	H	H	T	H

unobserved!

2

B
A
A
B
A

Coin A	Coin B
	5 H, 5 T
9 H, 1 T	
8 H, 2 T	
	4 H, 6 T
7 H, 3 T	
H:24, T:6	**H:9, T:11**

$$\hat{\theta}_A^0 = 0.60$$
$$\hat{\theta}_B^0 = 0.50$$

1

$$\hat{\theta}_A^1 \approx \frac{24}{24+6} = 0.80$$

$$\hat{\theta}_B^1 \approx \frac{9}{9+11} = 0.45$$

3

4

$$\hat{\theta}_A^2 \cong 0.80$$

$$\hat{\theta}_B^2 \cong 0.45$$

Case 3: Coin Unobserved, Soft Assignment

- **Soft assignment: a refinement on the naïve approach**
- **EM algorithm (expectation maximization)**
  - ***Expectation-step*** (using current parameters $\hat{\theta}^t$)
    - Computes **probabilities** for **each possible completion** of missing data

      These are used to create **weighted training examples**

*Expectation? $E[f(X)] = \sum_{x_i \in X} f(x_i) \cdot P(x_i)$

  - ***Maximization-step*** (using current latent variable values)
    - We do **Maximum** Likelihood Estimation to get a new (updated) parameter set $\hat{\theta}^{t+1}$

- ***Iterate E-step and M-step until converge***

Case 3: Coin Unobserved, Soft Assignment

☐ Example:

B	H	T	T	T	H	H	T	H	T	H	**5 H, 5 T**
A	H	H	H	H	T	H	H	H	H	H	**9 H, 1 T**
A	H	T	H	H	H	H	H	H	T	H	**8 H, 2 T**
B	H	T	H	T	T	T	H	H	T	T	**4 H, 6 T**
A	T	H	H	H	T	H	H	H	T	H	**7 H, 3 T**

unobserved!

$\hat{\theta}_A^0 = 0.60$
$\hat{\theta}_B^0 = 0.50$

1

E-step 2

A: 0.45	B: 0.55
A: 0.80	B: 0.20
A: 0.73	B: 0.27
A: 0.35	B: 0.65
A: 0.65	B: 0.35

weighted training examples

Head		Tail	
A: 2.2	B: 2.8	A: 2.8	B: 2.8
A: 7.2	B: 1.8	A: 0.8	B: 0.2
A: 5.9	B: 2.1	A: 1.5	B: 0.5
A: 1.4	B: 2.6	A: 2.1	B: 3.9
A: 4.5	B: 2.5	A: 1.9	B: 1.1
A: 21.3	**B: 11.7**	**A: 8.6**	**B: 8.4**

M-step 3

$$\hat{\theta}_A^1 \cong \frac{21.3}{21.3 + 8.6} \cong 0.71$$

$$\hat{\theta}_B^1 \cong \frac{11.7}{11.7 + 8.4} \cong 0.58$$

4

$\hat{\theta}_A^{10} \cong 0.80$
$\hat{\theta}_B^{10} \cong 0.52$

E-Step

- **Use current parameters $\hat{\theta}^t$ to compute a probability distribution over possible observations for $\mathbf{z} = \{z_1, \dots, z_5\}$, $z_i \in \{A, B\}$**
  - That is, we compute the posterior distribution: $p(z_i \mid x_i, \hat{\theta}^t)$
  - We know that $p(z_i \mid x_i, \hat{\theta}^t) = p(x_i \mid z_i, \hat{\theta}^t) \cdot p(z_i \mid \hat{\theta}^t) / p(x_i \mid \hat{\theta}^t)$

    assume [0.5, 0.5] not important (will be removed)

- **Compute $p_A = p(z_i = A \mid x_i, \hat{\theta}^t)$ and $p_B = p(z_i = B \mid x_i, \hat{\theta}^t)$**

H T T T H H T H T H $\qquad \hat{\theta}_A^0 = 0.60, \qquad \hat{\theta}_B^0 = 0.50$

$$p(x_1 = 5 \mid z_1 = A, \hat{\theta}_A^0) = 0.6^5 (1 - 0.6)^5 \cong 7.96 \times 10^{-4}$$
$$p(x_1 = 5 \mid z_1 = B, \hat{\theta}_B^0) = 0.5^5 (1 - 0.5)^5 \cong 9.77 \times 10^{-4}$$

$$\frac{P_A}{P_A + P_B} \cong 0.45 \quad (\text{coin } A)$$

$$\frac{P_B}{P_A + P_B} \cong 0.55 \quad (\text{coin } B)$$

E-Step => M-Step

Create weighted training examples

- It can be seen as **expected numbers of heads for A and B**

$$\frac{P_A}{P_A + P_B} \cong 0.45 \ \rightarrow 0.45 \times 5 \cong 2.2 \qquad (\text{coin } A)$$

$$\frac{P_B}{P_A + P_B} \cong 0.55 \ \rightarrow 0.55 \times 5 \cong 2.8 \qquad (\text{coin } B)$$

M-steps: do MLE using current weighted training examples

$$\hat{\theta}_A^1 = \frac{21.3}{21.3 + 8.6} \cong 0.76$$

$$\hat{\theta}_B^1 = \frac{11.7}{11.7 + 8.4} \cong 0.58$$

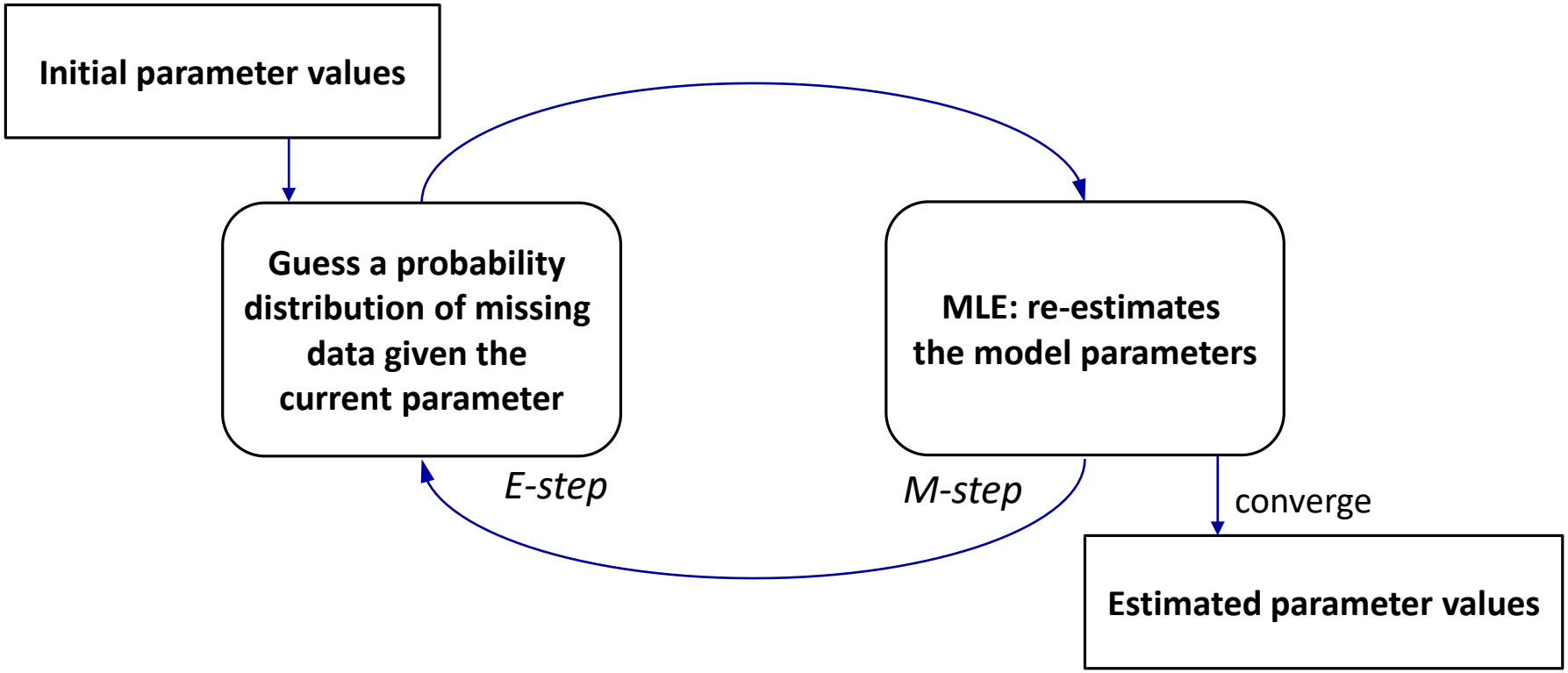
which will be used for computing the updated posterior distribution

$$p\left(z_i \middle| x_i, \hat{\theta}^1\right)$$

Summary

EM algorithm for latent variable models

- **E-step:** guesses a probability distribution over completions of missing data give current model parameters
- **M-step:** re-estimates model parameters using these completions via MLE

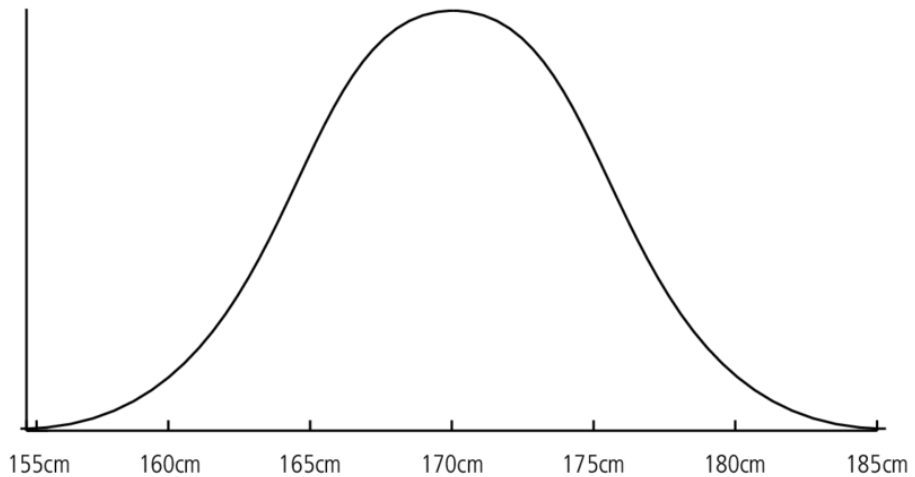


Recall: Gaussian PDF

Univariate Gaussian PDF

- Parameter $\boldsymbol{\theta} = (\mu, \sigma^2)$

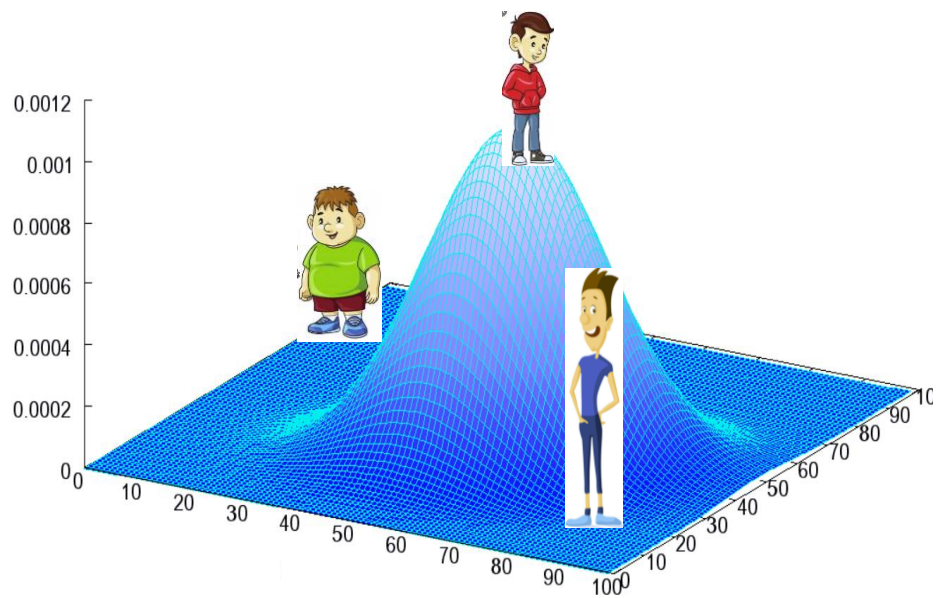
$$N(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$



Multivariate Gaussian PDF

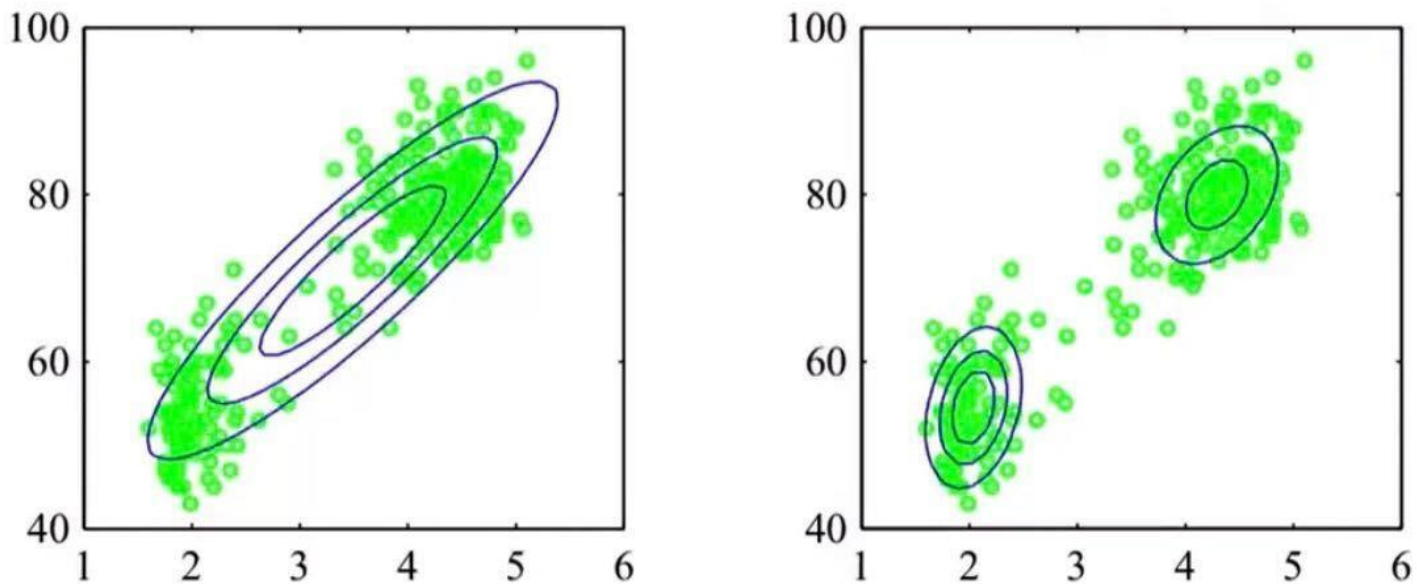
- Parameter $\boldsymbol{\theta} = (\boldsymbol{\mu}, \boldsymbol{\Sigma})$

$$N(\mathbf{x} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{|\sqrt{2\pi\boldsymbol{\Sigma}}|} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\Sigma})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})}$$



Motivation

- **Sometimes we need a linear *superposition* of multiple Gaussians**
  - Consider the following plots of '*math*' and '*natural sciences*' exam data: scores of 272 students (x-axis: score of '*math*'; y-axis: score of '*natural sciences*')
  - There might be two groups of students (but unobserved): 'sciences &engineering' group and '*non-sciences&engineering*' group



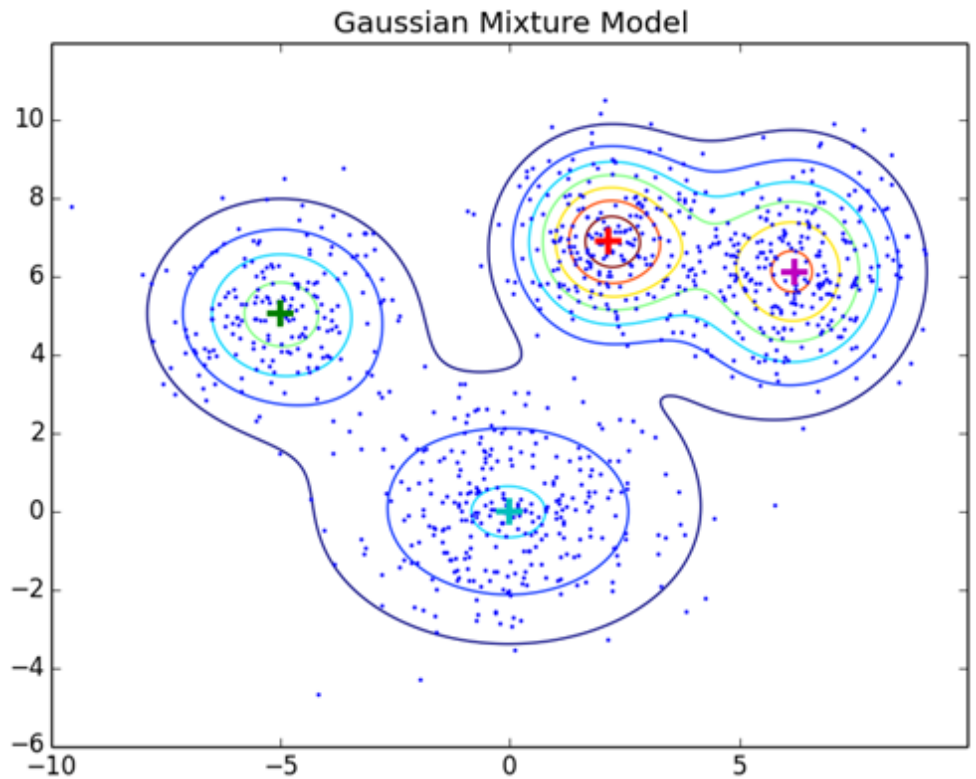
(a) simple (b) linear superposition

Gaussian Mixture Model

- **We need to mix K base Gaussian PDFs together**

$$p(\mathbf{x}) = \sum_{k=1}^{K} \pi_k \cdot N(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$

  - $N(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$: k^{th} base Gaussian for observations



- **Mixing weights π_k should be:**

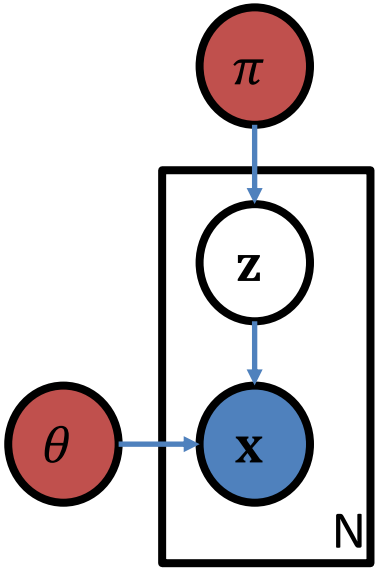
$$\sum_{k=1}^{K} \pi_k = 1, \qquad 0 \le \pi_k \le 1$$

EM Algorithm for GMM

- **Assume there are K components, then $p(\mathbf{x})$ is given by**

$$P(\mathbf{x}) = \sum_{k=1}^{K} P(z_k)P(x|z_k) = \sum_{k=1}^{K} \pi_k N(x|\mu_k, \Sigma_k)$$

  - Latent variable: $z_{nk} \in \{0,1\}$ for all data points
  - For simplicity, $\theta = (\mu, \Sigma)$ => $N(x|\mu_k, \Sigma_k) = N(x|\theta_k)$



- **Goal: to estimate $\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}$**
- **E-step: compute assignment probability**
  - $\gamma(z_{nk}) \equiv p(z_k = 1|x_n, \theta) = \frac{P(x_n|z_k = 1, \theta)P(z_k=1|\theta)}{\sum_{j=1}^{K} P(x_n|z_j = 1, \theta)P(z_j=1|\theta)}$

$$= \frac{\pi_k N(x_n|\theta_k)}{\sum_{j=1}^{K} \pi_j N(x_n|\theta_j)}$$

EM Algorithm for GMM

- **M-step: the computed $\gamma(z_{nk})$ are used to update π, μ, Σ**
  - The new $\gamma(z_{nk})$ motivates the update of the old parameters

- **Likelihood of the entire dataset is:**
  - $P(\mathbf{x}|\pi,\theta) \Rightarrow \prod_{n=1}^{N}\left(\sum_{k=1}^{K}\pi_k \cdot N(x_n|\theta_k)\right)$

$\Rightarrow \ln P(\mathbf{x}|\pi,\theta) \Rightarrow \sum_{n=1}^{N}\ln\{\sum_{k=1}^{K}\pi_k \cdot N(x_n|\theta_k)\}$

$\Rightarrow \underset{\theta}{\operatorname{argmax}} \ln P(\mathbf{x}|\pi,\theta) = \underset{\theta}{\operatorname{argmax}} \sum_{n=1}^{N}\ln\{\sum_{k=1}^{K}\pi_k \cdot N(x_n|\theta_k)\}$

- **Parameter update rules:**
  - $\frac{d}{d\mu_k}\ln P(\mathbf{x}|\pi,\mu,\Sigma) = \sum_{n=1}^{N}\frac{\pi_k N(x|\mu_k,\Sigma_k)}{\sum_{j=1}^{K}\pi_j N\left(x\middle|\mu_j,\Sigma_j\right)}\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}_n - \hat{\mu}_k) = 0$

$$\rightarrow \sum_{n=1}^{N}\gamma(z_{nk})\,(\boldsymbol{x}_n - \hat{\mu}_k) = 0 \quad \rightarrow \boxed{\hat{\mu}_k = \frac{\sum_{n=1}^{N}\gamma(z_{nk})\,\boldsymbol{x}_n}{\sum_{n=1}^{N}\gamma(z_{nk})}}$$

EM Algorithm for GMM

Parameter update rules:

- $\frac{d}{d\Sigma_k}\ln P(X|\pi,\mu,\Sigma) = 0$

$$\rightarrow \boxed{\Sigma_k = \frac{\sum_{n=1}^{N}\gamma(z_{nk})\,(\boldsymbol{x}_n - \hat{\mu}_k)(\boldsymbol{x}_n - \hat{\mu}_k)^{\boldsymbol{T}}}{\sum_{n=1}^{N}\gamma(z_{nk})}}$$

- $\frac{d}{d\pi_k}\ln P(X|\pi,\mu,\Sigma) + \lambda\left(\sum_{k=1}^{K}\pi_k - 1\right) = 0$ (**Lagrange method** is used here)

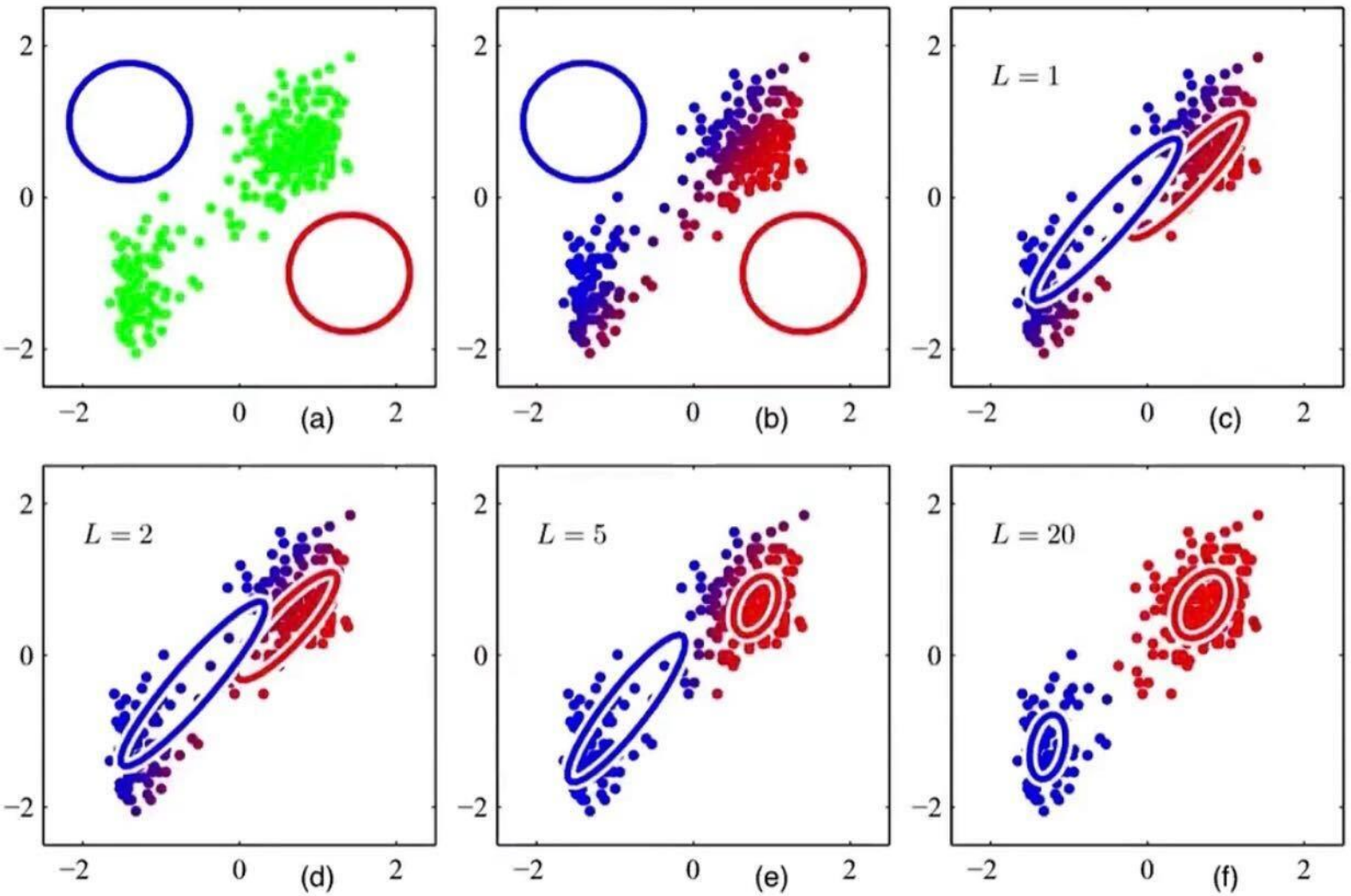
$$\rightarrow \sum_{n=1}^{N}\frac{N(x|\mu_k,\Sigma_k)}{\sum_{j=1}^{K}\pi_j N\left(x\middle|\mu_j,\Sigma_j\right)} + \lambda = 0$$

$$\rightarrow \sum_{k=1}^{K}\left\{\sum_{n=1}^{N}\frac{\pi_k N(x|\mu_k,\Sigma_k)}{\sum_{j=1}^{K}\pi_j N\left(x\middle|\mu_j,\Sigma_j\right)} + \pi_k\lambda\right\} = 0$$

$$\rightarrow \lambda = -N \rightarrow \boxed{\pi_k = \frac{\sum_{n=1}^{N}\gamma(z_{nk})}{N}}$$

GMM Example

- Updates by iterations (K=2)



EM Algorithm for GMM (Summary)

- **Run EM**
  - Because two (observed and hidden) variables are interacting
- **Expectation step**
  - Compute assignment probability of each observation to each Gaussian
  - $\gamma(z_{nk}) \equiv p(z_k = 1|x_n) = \frac{P(x_n|z_k = 1, \theta)P(z_k=1|\theta)}{\sum_{j=1}^{K} P\left(x_n\middle|z_j = 1, \theta\right)P(z_j=1|\theta)}$

$$= \frac{\pi_k N(x_n|\theta_k)}{\sum_{j=1}^{K} \pi_j N\left(x_n\middle|\theta_j\right)}$$

- **Maximization step**
  - Update the parameters given $\gamma(z_{nk})$

$$\hat{\mu}_k = \frac{\sum_{n=1}^{N} \gamma(z_{nk})\, x_n}{\sum_{n=1}^{N} \gamma(z_{nk})} \qquad \hat{\Sigma}_k = \frac{\sum_{n=1}^{N} \gamma(z_{nk})\, (\boldsymbol{x}_n - \hat{\mu}_k)(\boldsymbol{x}_n - \hat{\mu}_k)^{\boldsymbol{T}}}{\sum_{n=1}^{N} \gamma(z_{nk})} \qquad \hat{\pi}_k = \frac{\sum_{n=1}^{N} \gamma(z_{nk})}{N}$$

Hard Assignment: K-Means Clustering

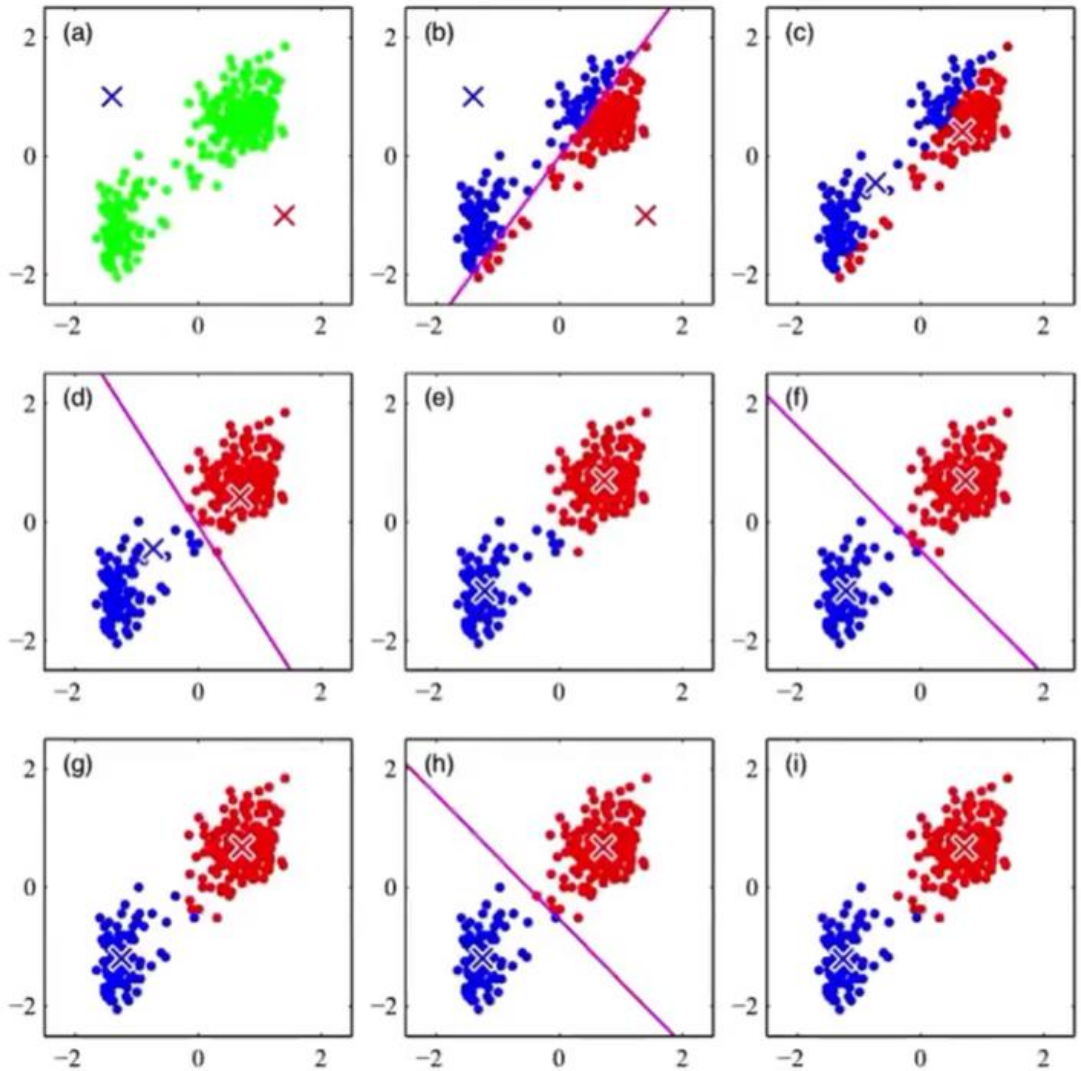
- **It is a mixture model without probability theory**
- **Algorithm**
  - Setup K number of centroids
    - $\boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \dots, \boldsymbol{\mu}_K$
  - Cluster data points by the distance from the points to the **nearest** centroid
- **Formally,**
  - The loss function $J = \sum_{n=1}^{N} \sum_{k=1}^{K} z_{nk} ||\mathbf{x}_n - \boldsymbol{\mu}_k||^2$
  - Minimize J by optimizing
    - z_{nk}: the assignment of data points to clusters
    - $z_{nk} \in \{0, 1\}$ with $k = 1, 2, 3, \dots, K$
    - $\boldsymbol{\mu}_k$: the location of centroids
- **EM (==iterative optimization) is used**
  - Fix one, and optimize the other, repeat this until converge

K-Means Clustering

- **Loss function:** $J = \sum_{n=1}^{N} \sum_{k=1}^{K} z_{nk} ||\mathbf{x}_n - \boldsymbol{\mu}_k||^2$
  - Expectation step
    - Assign (z_{nk}) the data points to the nearest centroid
    - Logical choice: the nearest centroid $\boldsymbol{\mu}_k$ for a data point of $\mathbf{x}_n$
  - Maximization step
    - Update the centroid positions ($\boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \dots, \boldsymbol{\mu}_K$) given the assignments
    - $\frac{dJ}{d\boldsymbol{\mu}_k} = \frac{d}{d\boldsymbol{\mu}_k} \sum_{n=1}^{N} \sum_{k=1}^{K} z_{nk} ||\mathbf{x}_n - \boldsymbol{\mu}_k||^2 = 0 \quad \Rightarrow \quad \boldsymbol{\mu}_k = \frac{\sum_{n=1}^{N} z_{nk} \mathbf{x}_n}{\sum_{n=1}^{N} z_{nk}}$

Example

☐ Updates by iterations (K=2)



EM Algorithm in General

Preliminary

- X: observed variables
- Z: hidden (latent) variables
- θ: parameters for distribution of X
- Likelihood $P(X|\theta) = \sum_Z P(X, Z|\theta)$

Log-likelihood → $\ln P(X|\theta) = \ln\{\sum_Z P(X, Z|\theta)\}$

Goal

- The values of Z and θ
  - Optimizing $P(X|\theta) = \sum_Z P(X, Z|\theta)$
  - EM => do MLE together with considering Z

Maximizing the Lower Bound

Use the Jensen's inequality

can be any type of probability distribution over Z

- $\ln P(X|\theta) = \ln\{\sum_Z P(X,Z|\theta)\} = \ln\{\sum_Z q(Z)\frac{P(X,Z|\theta)}{q(Z)}\}$

- Jensen's inequality:

  When $\varphi(x)$ is concave
  $\varphi(\mathrm{E}[X]) \geq \mathrm{E}[\varphi(X)]$

  When $\varphi(x)$ is convex
  $\varphi(\mathrm{E}[X]) \leq \mathrm{E}[\varphi(X)]$

- Thus, $\ln\{\sum_Z q(Z)\frac{P(X,Z|\theta)}{q(Z)}\} \geq \sum_Z q(Z)\ln\frac{P(X,Z|\theta)}{q(Z)}$

Now our goal becomes to maximize the lower bound

- $\ln P(X|\theta) \geq \boxed{\sum_Z q(Z)\ln\frac{P(X,Z|\theta)}{q(Z)}} = \sum_Z q(Z)\ln\frac{P(Z|X,\theta)P(X|\theta)}{q(Z)}$

$$= \sum_Z \{q(Z)\ln P(X|\theta) + q(Z)\ln\frac{P(Z|X,\theta)}{q(Z)}\}$$

$$= \ln P(X|\theta) + \sum_Z \{q(Z)\ln\frac{P(Z|X,\theta)}{q(Z)}\}$$

KL-Divergence

- **We just derived a term that makes difference**
  - $\ln P(X|\theta) \geq \ln P(X|\theta) - \sum_Z \{q(Z) \ln \frac{q(Z)}{P(Z|X,\theta)}\}$

- **The second term is a form of KL-divergence**
  - Kullback−Leiber divergence, or KL−divergence measures the **difference between two probability distributions P and Q**
    - $KL(P||Q) \geq 0$
    - If $P == Q$, $KL(P||Q) = 0$
  - KL divergence $KL(P||Q) = \sum_i P(i) \ln(\frac{P(i)}{Q(i)})$
  - $KL(q(Z)\,||P(Z|X,\theta)) = \sum_Z \{q(Z) \ln \frac{q(Z)}{P(Z|X,\theta)}\}$

Maximizing the Lower Bound

- **Our new objective: to minimize the KL-divergence term**
  - $\ln P(X|\theta) \geq \ln P(X|\theta) - \sum_Z \{q(Z) \ln \frac{q(Z)}{P(Z|X,\theta)}\}$
    - How to set $q(Z)$ to minimize the KL-divergence term?
    - $KL(q(Z)||P(Z|X,\theta)) = 0 \rightarrow q(Z) = P(Z|X,\theta)$
  - Recall: the **lower bound** to be maximized is:
    - $\sum_Z q(Z) \ln \frac{P(X,Z|\theta)}{q(Z)} = \sum_Z q(Z) \ln P(X,Z|\theta) - q(Z) \ln q(Z)$

$$= \sum_Z q(Z) \ln P(X,Z|\theta) + H(q)$$

H: entropy, will be ignored

    - Let current lower bound function $Q(\theta^t, q) = \sum_Z q(Z) \ln P(X,Z|\theta) + H(q)$

Of course, the lower bound $Q(\theta^t, q) = \ln P(X|\theta) - KL(q(Z)\,||P(Z|X,\theta))$

EM Algorithm

- **EM Finds the MLE solutions for models with latent variables**
  - $P(X|\theta) = \sum_Z P(X,Z|\theta) \rightarrow \ln P(X|\theta) = \ln\{\sum_Z P(X,Z|\theta)\}$
- **Algorithm**
  - Initialize θ^0 randomly
  - Loop until the likelihood converges
    - **Expectation step: how to setup Z for each time t?**
      - Lower bound: $Q(\theta, q^t)$
      - $q^{t+1}(Z) = \text{argmax}_q Q(\theta^t, q) = \text{argmin}_q KL(q(Z)||P(Z|X,\theta^t))$

      => $q^t(Z) = P(Z|X,\theta^t)$: Assign Z by following the distribution $P(Z|X,\theta^t)$
      - $q^t(Z)$ means the distribution of latent variable Z at time t

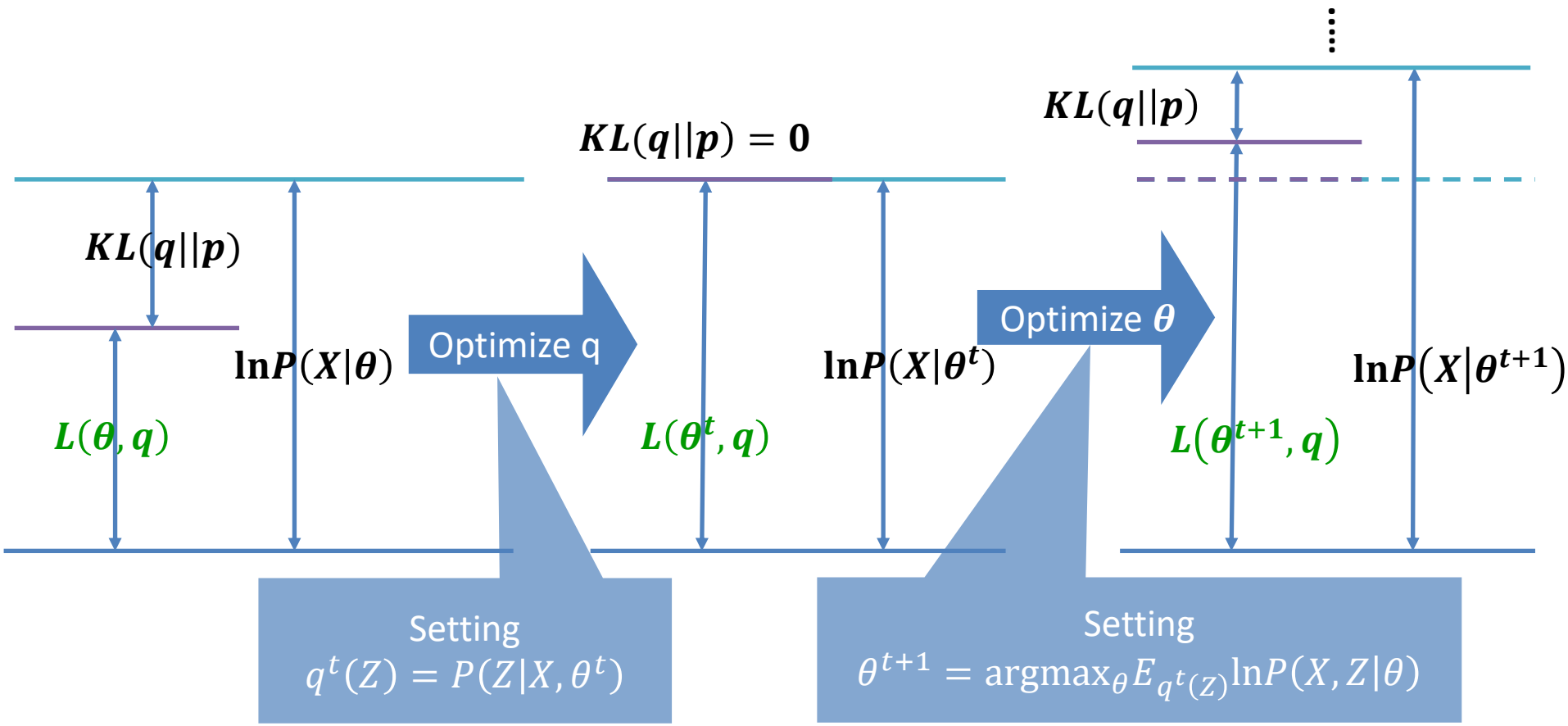
    - **Maximization step**
      - $\hat{\theta}^{t+1} = \underset{\theta}{\text{argmax}} Q(\theta, q^t) = \underset{\theta}{\text{argmax}} \sum_Z q(Z) \ln P(X,Z|\theta)$

        $= \underset{\theta}{\text{argmax}} \sum_Z p(Z|X,\theta^t) \ln p(X,Z|\theta)$

EM Algorithm

Graphical illustrations

- Let $\ln P(X|\theta) = L(\theta, q) + \sum_Z \{q(Z) \ln \frac{q(Z)}{P(Z|X,\theta)}\}$



Re-visit GMM

- **EM update rule (in general):**

$$\theta^{t+1} = \arg\max_{\theta} \sum_{Z} \underbrace{p(Z|X,\theta^t)}_{\text{Probability over latent}} \underbrace{\ln p(X,Z|\theta)}_{\text{Likelihood with Z assigned}}$$

  - Each completion: weighted by the posterior probability
  - **Maximizes the log likelihood** of weighted training examples

- **In GMM,**

$$\theta^{t+1} = \underset{\theta}{\operatorname{argmax}} \ln\{\sum_{k=1}^{K} \overset{\text{Probability over latent}}{\pi_k} \underbrace{N(X|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}_{\text{Likelihood with } k \text{ assigned}}\}$$

thus, we computed

$$\gamma(z_{nk}) \equiv p(z_k = 1|x_n) = \frac{P(x_n|z_k = 1,\theta)P(z_k=1|\theta)}{\sum_{j=1}^{K} P\left(x_n\middle|z_j = 1,\theta\right)P\left(z_j=1|\theta\right)}$$

$$= \frac{\pi_k N(x_n|\theta_k)}{\sum_{j=1}^{K} \pi_j N\left(x_n\middle|\theta_j\right)}$$

Re-visit GMM

- **And then the parameter update rules are established based on $\gamma(z_{nk})$:**
  - $\frac{d}{d\mu_k}\ln P(\mathbf{x}|\pi,\mu,\Sigma) = 0 \quad \Rightarrow \quad \hat{\mu}_k = \frac{\sum_{n=1}^{N}\gamma(z_{nk})x_n}{\sum_{n=1}^{N}\gamma(z_{nk})}$
  - $\frac{d}{d\Sigma_k}\ln P(\mathbf{x}|\pi,\mu,\Sigma) = 0 \quad \Rightarrow \quad \hat{\Sigma}_k = \frac{\sum_{n=1}^{N}\gamma(z_{nk})(\boldsymbol{x}_n-\hat{\mu}_k)(\boldsymbol{x}_n-\hat{\mu}_k)^{\boldsymbol{T}}}{\sum_{n=1}^{N}\gamma(z_{nk})}$
  - $\frac{d}{d\pi_k}\ln P(\mathbf{x}|\pi,\mu,\Sigma) + \lambda\left(\sum_{k=1}^{K}\pi_k - 1\right) = 0 \quad \Rightarrow \quad \hat{\pi}_k = \frac{\sum_{n=1}^{N}\gamma(z_{nk})}{N}$

Thank You !